

HOUSE PRICE PREDICTION SYSTEM

Using Machine Learning and Regression

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Project Type: Minor Project

Domain: Artificial Intelligence / Machine Learning

Platform: Google Colab

Submission Deadline: 13 August 2026

Problem Statement

Accurately estimating house prices is challenging due to multiple influencing factors such as property size, quality, location, age, rooms, garage facilities, and other characteristics. A machine learning system can learn patterns from historical housing data and use these patterns to estimate the sale price of a property.

Project Overview

The House Price Prediction System is a machine learning project designed to predict residential property sale prices using property-related features. The project involves cleaning and preprocessing housing data, performing feature engineering, training regression models, and evaluating prediction accuracy. The system provides hands-on experience with structured data prediction problems.

Objectives

- Clean and preprocess housing data.
- Analyze important property-related features.
- Perform feature engineering.
- Prepare numerical and categorical features for machine learning.
- Train regression models to predict house prices.
- Evaluate prediction accuracy using regression metrics.
- Compare model performance and identify a suitable model.

Dataset Information

Dataset Name: House Prices Dataset

Source: Kaggle

Link: <https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/data>

The House Prices dataset contains property-related attributes and a SalePrice target in the training data. These features are used to build a regression model capable of estimating house prices.

Tools & Technologies Used

- Python
- Google Colab
- Pandas
- NumPy
- Matplotlib
- Scikit-learn

AI Techniques Used

- Regression
- Feature engineering
- Data preprocessing
- Categorical feature encoding
- Model evaluation

System Workflow

Housing Dataset → Data Cleaning → Exploratory Data Analysis → Feature Engineering → Preprocessing → Regression Model → Price Prediction → Model Evaluation

Methodology

1. Data Loading

The housing dataset is loaded into a Pandas DataFrame in Google Colab. The dataset is inspected to understand its size, columns, data types, and target variable.

2. Data Cleaning

Missing values, inconsistent entries, and unsuitable data types are identified and handled before model training.

3. Exploratory Data Analysis

Statistical summaries and visualizations are used to understand the relationship between property features and house sale prices.

4. Feature Engineering

Useful property-related features can be created or transformed from existing columns to improve the model's ability to learn relationships in the data.

5. Feature Preprocessing

Numerical and categorical variables are prepared for machine learning. Categorical features can be encoded into numerical representations.

6. Model Training

Regression algorithms are trained using the processed property features and SalePrice as the target variable.

7. Model Evaluation

Prediction accuracy is evaluated using appropriate regression metrics such as MAE, MSE, RMSE, and R^2 .

8. Price Prediction

The trained model is used to predict house prices for unseen property records.

Implementation / Work Evidence

The following screenshots should be inserted into the final submitted report after completing the Google Colab implementation:

- Google Colab notebook showing library imports.
- Dataset loading and DataFrame preview.
- Dataset shape, columns, and data types.
- Missing-value analysis.
- Exploratory data analysis graphs.
- Feature engineering code and output.
- Preprocessing and categorical encoding.
- Regression model training.
- Model evaluation metrics.
- Prediction output.
- Actual-vs-predicted or other model performance visualization.

Expected Output

The completed system should generate predicted house prices and a model evaluation summary. Actual metric values should be inserted from the final Google Colab run rather than being estimated in the report.

Model	Evaluation Metric	Result
Linear Regression	RMSE / R^2	Add Colab result
Random Forest Regressor	RMSE / R^2	Add Colab result
Best Model	RMSE / R^2	Add final result

Screenshots / Work Evidence

Insert the actual Google Colab screenshots here. The screenshots should demonstrate data preprocessing, feature engineering, model training, evaluation, and final prediction output.

Conclusion

The House Price Prediction System demonstrates how machine learning can be applied to a structured data prediction problem. By cleaning housing data, performing feature engineering, training regression models, and evaluating their performance, the project provides practical experience in the complete machine learning workflow. The system can use property-related features to estimate residential sale prices.

Future Scope

- Compare additional regression and ensemble algorithms.
- Perform systematic hyperparameter tuning.
- Improve feature selection and feature engineering.
- Develop an interactive house-price prediction interface.
- Deploy the trained model as a web application.
- Add model explainability to identify important price-influencing features.

External Link

GitHub Repository: <https://github.com/Eshaan1234/House-Price-Prediction-System-Minor-Project-Internship>

Dataset: <https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/data>

References

Kaggle – House Prices: Advanced Regression Techniques.

Pandas documentation.

Matplotlib documentation.

Scikit-learn documentation.